

MANTHAN CHAWDA

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EDUCATION

GSFC University

B.Tech in Computer Science & Engineering / CGPA: 8.47

Bright School

Higher Secondary & Secondary Education, PCM

Vadodara, Gujarat

July 2024 – Present

Vadodara, Gujarat

May 2024

EXPERIENCE

Academic Associate

GSFC University

January 2026 – Present

Vadodara, Gujarat

- Guide undergraduate students through experimental physics laboratory sessions and help bridge the gap between theory and practical application.
- Set up and maintain laboratory equipment and supervise hands-on experiments while ensuring proper equipment handling and laboratory safety.

Intern

GUIITAR Council, GSFC University

December 2025

Vadodara, Gujarat

- Researched aerodynamic and modular drone designs inspired by mini-FPV platforms with a focus on improving flight efficiency and range.
- Analyzed and proposed structural design iterations for lightweight 3D-printed wings and body components to improve aerodynamics and flight duration.

PROJECTS

CivicPulse | YOLO11m-seg, ByteTrack, MiDaS, Python, PyTorch, FastAPI

August 2026 – Present

- Developed an AI-powered civic infrastructure monitoring platform that analyses drone footage to detect potholes, waterlogging, open manholes, drainage overflow and damaged footpaths.
- Built a computer vision and incident-analysis pipeline using YOLO11m-seg, ByteTrack and MiDaS and integrated it with FastAPI, React/Vite and Supabase/PostgreSQL for telemetry-aware dashboard monitoring.

ClearContract | Gemini API, React, Chrome Extension, Node.js, Express, MongoDB

July 2026 – Present

- Developed a Chrome extension prototype that analyses pasted contract and Terms of Service text using Google's Gemini API to identify potentially concerning clauses and explain them in simpler language.
- Built the extension-to-backend workflow using Node.js and Express with MongoDB-based application components and authentication while continuing to develop the dashboard.

Meshmerise | C++, Arduino IDE, STM32, RLS08, TB6612FNG

October 2025 – December 2025

- Designed and built an autonomous maze-solving robot using an STM32 microcontroller and RLS08 line sensors for navigation.
- Implemented line-following and left-hand maze-solving logic in Arduino IDE and optimized second-run traversal using recorded turn sequences.

ACHIEVEMENTS

First Runner-Up – ELCIA Next-Gen Innovative Tech Hackathon 2026

Smart City Drone-AI Challenge – Team CivicPulse, among 183 teams from 18 states and 100+ institutions.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Swift, HTML/CSS

AI/ML: PyTorch, TensorFlow, Scikit-learn, OpenCV, NumPy, pandas, Matplotlib

Frameworks: FastAPI, React, Node.js, Express

Databases & Tools: PostgreSQL, MongoDB, Git, Docker, VS Code, Arduino IDE, Fusion 360

CERTIFICATIONS

Getting Started with Artificial Intelligence – IBM

August 2026

Networking Basics – Cisco Networking Academy

March 2026

TIDES Ignition Program 2026 – IIT Roorkee

January 2026

Oracle Cloud Infrastructure Data Science Professional

October 2025

Oracle Cloud Infrastructure AI Foundations

October 2025

Python Essentials 1 & 2 – Cisco Networking Academy

September 2025

AWS Cloud Foundations – Amazon Web Services

July 2025